

Prediction of bleeding behavior and film thickness evolution in grease lubricated rolling contacts

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ABSTRACT

While a rolling element bearing is operating, the lubricating film is desired to maintain a sufficient thickness to avoid asperity contact between surfaces. When using grease as lubricant, the film thickness undergoes evolution over time, due to the transient formation of the deposited thickener-rich layer and a bled oil film. Therefore, a model, which can address both type of film formation, is required. In this contribution, an experimentally validated numerical model is proposed by assuming the grease as a porous medium. This model can calculate the transient formation of thickener-rich layer and the oil bleeding process for grease-lubricated contacts. The predicted film thickness evolution, by superposing the thickener-rich layer and oil film, shows a reasonable agreement with the experimental result.

1. Introduction

To avoid direct asperity contact and severe friction losses during the operation of rolling element bearings, the lubricating films between contacting surfaces are desired to maintain sufficient thickness [1]. Therefore, the understanding and prediction of the lubricating film formation for a lubricant is particularly important. Nowadays, about 90% of rolling element bearings are lubricated using greases, which consist of base oil, additives, and thickener [2]. Due to its porous microstructure [3–5], the thickener can retain oil and provide the grease with a specific consistency [4]. After grease is applied to a rolling element bearing, the fresh grease is being entrained into the contact [6] and large thickener clumps can be observed during the initial churning phase [7]. The grease that has passed through the contact, is degraded by the shear effects in the contact zone, becoming a thickener/oil glomerates [8]. The oil can be expelled by the contact pressure, while the thickener is left, leading to the formation of a solid-like thickener-rich layer on the contact surface [6,9,10]. Over time, the bulk grease is displaced from the contact zone by the rolling elements [11]. Due to its consistency, the displaced grease cannot flow back leading to a depletion of grease in the running track and the formation of grease reservoirs on both sides of the track [12,13], as schematically shown in Fig. 1. After the churning phase, the bleeding phase starts [14]. A limited amount of oil is slowly bleeding out from these reservoirs during the repetitive passes of the rolling elements, leading to an enhanced

elasto-hydrodynamic lubrication (EHL) oil film formation [13].

Due to the aforementioned effects, the lubricating film thickness in the bleeding phase undergoes an evolution over time, even under unchanged operating conditions [8,13,16,17]. When the amount of bled oil is sufficient, such that the oil film thickness is not limited by the amount of oil supplied to the contact inlet, the lubrication condition is termed fully flooded. Conversely, the condition with an insufficient oil supply is referred to as starvation [14]. Generally, four characteristic film thickness evolutions can be identified: fully flooded, starved with stabilization, monotonically starved and starved with recovery [13,17,18]. The fully flooded film evolution requires a high bleeding rate, and therefore results in a shorter service life of the grease [17,18]. In contrast, at an insufficient bleeding rate, the oil film cannot be formed efficiently [19–22], such that the grease film is reduced to a low thickness with stabilization and mainly consists of a thickener-rich layer [23]. The thickener-rich layer is independent of the rolling speed [8,23], therefore has important implications for start-stop operations [9] or low rolling speed conditions [10]. However, the formation of this layer depends on the bulk grease flow in the churning phase [8] and can be consumed by shear effects [9,24]. When the thickener-rich layer is insufficient to ensure the contact separation, a monotonically starved evolution occurs [17,24]. Therefore, for the long-term operation of grease-lubricated bearings, the ideal thickness evolution is the starved film thickness with recovery [9], which requires a proper bleeding rate [11,13,23,25]. Besides, the formation of oil film is affected by the thickener-rich layer because it can change the wettability between oil and the contacting

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Nomenclature*Latin characters*

a, b	Hertzian radius [m]
Ca	Capillary number [-]
E	Elastic modulus [Pa]
f	Thickener fraction [-]
F_{aim}	Aim force [N]
g	Gravitational constant [$m \cdot s^{-2}$]
h	Film thickness [nm]
h_{Λ}	Critical film thickness [nm]
k	Permeability [m^2]
L_{track}	Length of the track
n	Number of disc revolution [-]
N	Max. number of disc revolution [-]
p	Pressure [Pa]
p_c	Capillary pressure [Pa]
$\hat{q}$	Side flow rate [$m \cdot s^{-1}$]
r	Fiber radius [nm]
R_q	Surface Roughness [nm]
R	Radius [m]
S_{oil}	Oil saturation before infiltration [-]
$\tilde{S}_{oil}$	Oil saturation after infiltration [-]
SRR	Sliding-to-rolling ratio [-]
t	Time [s]
u, v, w	Rolling speed [$m \cdot s^{-1}$]
V	Mesh cell volume [$m^3 \cdot s^{-1}$]
w_{grease}	Width of the simulation domain [m]

Greek characters

α	Fiber tilting angle [°]
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β_{Barus}	Pressure-viscosity coefficient [GPa-1]
$\dot{\gamma}$	Shear rate [$m \cdot s^{-1} \cdot m^{-1}$]
δ	Surface normal displacement [m]
ϵ	Porosity [-]
η	Dynamic viscosity [Pa-s]
$\bar{\eta}$	Dimensionless viscosity [-]
θ	Contact angle [°]
ν	Poisson's ratio [-]
$\bar{\rho}$	Dimensionless density [-]
ρ	Density [$kg \cdot m^{-3}$]
τ_y	Yield stress [Pa]

Subscripts

c	Contact center
ff	Fully flooded condition
Hz	Hertzian condition
oil	Oil film
$p0$	Atmospheric pressure condition
th	Thickener-rich layer
s	Starved condition
$void$	Void space
$inter$	Interface between thickener-rich layer and oil film
x	x direction
y	y direction
z	z direction
$\perp$	Perpendicular to the direction of flow
$//$	Parallel to the direction of flow

Superscripts

$\sim$	free surface condition
$+, -$	Positive value, negative value

surfaces[24,26]. Consequently, when predicting the grease film evolution, both components, the thickener-rich layer and the oil film, should be addressed.

Due to the complexity of the two-component lubrication hypothesis [27], the grease film can be assumed as a superposition of two distinct components: the thickener-rich layer and the oil film [9,23]. Therefore, it is appropriate to model the formation of these components separately [9].

Firstly, the simulation of thickener-rich layer is still new and only a few models have direct relevance [9]. Recently, Tichy et al. [27] and

Nogi et al. [28] extended the porous medium concept to a line and point contact, respectively. The thickener-rich layer is modeled as a mixture of thickener and oil, and a porous medium where oil can flow through [27, 28]. The numerical results show an increase of thickener fraction in the track center, due to the side flow of the oil retained within the thickener structure [28]. The flow is enforced by the contact pressure [28]. However, as of today, only the grease flow under steady state in the rolling contact is modeled, and therefore the transient formation of a thickener-rich layer cannot be clarified.

Secondly, the prediction of oil film thickness is typically addressed

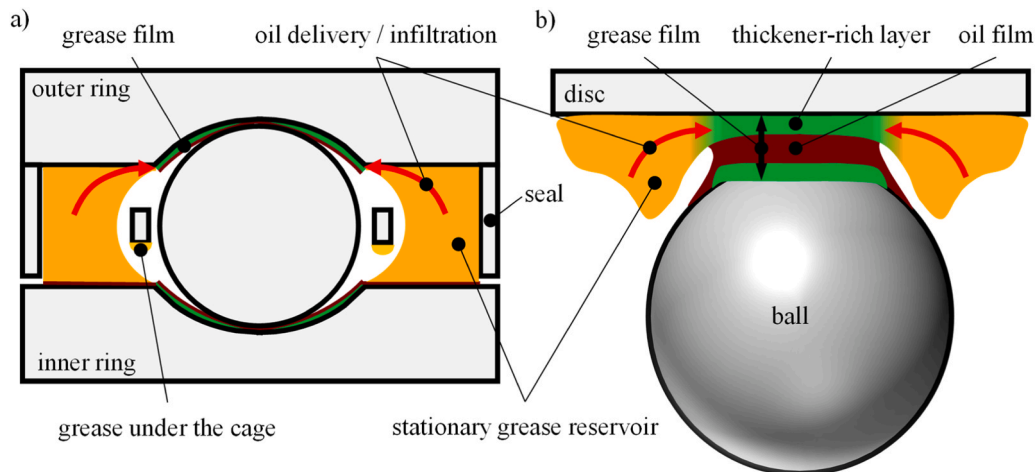


Fig. 1. Schematic representation of grease distribution after churning (along cross rolling direction view), (a) in a ball bearing with a rotating inner ring, reproduced from [15], (b) in a ball-on-disc contact.

using starvation models [14]. Under starvation, the oil is displaced to the sides of the running track and forms two side bands [26]. In front of the consecutive contact, the oil is replenished by the surface tension effects and forms an air/oil meniscus [21,22,29,30]. Hamrock and Dowson formulated a film thickness equation as a function of the meniscus position [31], which can be determined using CFD-simulations under the assumption that the available oil amount is known [21,22]. For grease lubrication under severe starvation, the meniscus touches the Hertzian contact circle [13,17]. As a results, this method becomes invalid [19,26,32]. As an alternative to the meniscus method, Cann et al. experimentally identified four primary factors which control the starvation degree (SD), including available oil amount, load, oil viscosity and speed [33]. Then, an empirical equation between the SD and the oil film thickness is established [33,34]. However, both methods need the oil amount as an input parameter to determine the oil film thickness.

To the best of the authors' knowledge, accurately measuring the bled oil amount for grease-lubricated rolling contacts in tribometer setups is challenging and nearly impossible in real applications. Therefore, Baart et al. developed an oil bleeding model on the basis of the porous medium assumption [4]. They assumed the centrifugal force to be the driving force of bleeding and used the Darcy's equation to calculate the bleeding rate [4]. The model is promising to evaluate the oil bleeding from grease reservoirs rotating with a high speed, such as the greases located at the cage or inner ring of a rolling element bearing [35]. However, after the churning phase, greases tend to settle at the sealings and the outer ring of a bearing [15,36,37], as shown in Fig. 1(a). The greases at both locations are free from the centrifugal force and can be considered as stationary [37]. As indicated by Komoriya et al., the stationary greases deliver oil through the thickener-rich layer to the contact because of a so-called infiltration effect [36], see Fig. 1. Therefore, the driving force of bleeding from the stationary grease is different from the centrifugal force in the aforementioned model [4].

To identify this driving force, Zhang et al. recently developed a grease-on-paper test configuration [15,37]. They demonstrate that the stationary grease reservoirs can bleed oil by the capillary pressure generated within the blotting paper [15,37]. The capillary pressure is defined as the pressure difference between air phase and oil phase [38], therefore it can drive the oil infiltrating the blotting paper. By assuming grease and blotting paper to have porous structures, they proposed a model to calculate the bled oil volume [39]. This testing method and model help to understand the capillary pressure driven oil flow in the vicinity of rolling contacts. However, Zhang et al. assumed the grease reservoirs as a cylindrical patch in their model [15,37]. This is in contrast to grease-lubricated rolling contacts, where the grease typically forms a track [22] (see Fig. 1), which potentially affects the bled oil flow. Furthermore, the thickener-rich layer, which is the pathway of oil infiltration in grease-lubricated contacts [36], is formed gradually due to the overrolling of the contact [6,9,24,40]. The porous properties of the thickener-rich layer, including the permeability and porosity, are time-dependent when oil infiltrates from grease reservoirs to the track center [25]. In contrast, the porous properties of the blotting paper in Zhang's model are constant [15,37]. Hence, the transient changes of permeability and porosity are not included in Zhang's model to determine the transient oil bleeding rate, which would be of particular importance to quantify the amount of bled oil during operation of a grease-lubricated contact. Consequently, the grease film thickness evolution cannot be predicted with the available models.

The aim of this study is to develop an experimentally validated numerical model, which can be used to predict the film thickness evolution for a grease-lubricated rolling contact in the bleeding phase. The grease film is assumed as the superposition of the thickener-rich layer and the oil film. The transient formation of thickener-rich layer is modeled by a porous thin layer model, using the porous concepts by Tichy et al. [27] and Nogi et al. [28]. And a multiphase porous medium model is adopted to simulate the oil bleeding. As a key factor influencing the grease lubrication, the effects of thickener on the film formation evolution are

studied.

2. Numerical models

2.1. Porous thin layer model

2.1.1. Flow model of thickener-rich layer in the contact

The thin layer model assumes that the rolling track is covered by a thin layer of lubricant [41], as shown in Fig. 2. This lubricant layer is continuously passed by the contacts, resulting in the lateral side flow of the lubricant (the z -direction in Fig. 2) [41]. Consequently, the central film thickness decreases as a result of mass conservation. According to Fischer et al. [23], the thickness of thickener-rich layer on both the disc and ball surfaces is nearly equal, as represented schematically by the green layers in the contact zone of Fig. 2(b). Therefore, the thickness of the thickener-rich layer (h_{th}) is assumed to be twice that of the thickener-rich layer on the disc. Only the layer on the disc surface is simulated in this study. The sensitivity check of this assumption can be found in the appendix Fig.A1, which shows the predicted thickness of the thickener-rich layer is not sensitive to this assumption.

To predict the thickness of the thin layer, van Zoelen et al. derived an equation based on the mass conservation of the lubricant in the contact zone [41]:

$$\frac{\partial[h_{th}(x, z, t)]}{\partial t} = -\frac{2}{\rho L_{track}} \frac{\partial \hat{q}_z}{\partial z} \quad (1)$$

where h_{th} is the thickness of thickener-rich layer, $\hat{q}_z$ is the side flow flux of the thickener-rich layer at z direction and L_{track} is the track length at the rolling direction. ρ is the mixture density of thickener and base oil. As an approximation, the Dowson and Higginson equation is used for the density-pressure behavior of both components in this work, which is similar to [28]:

$$\rho = \rho_{p0} \frac{5.9 \times 10^8 + 1.34p}{5.9 \times 10^8 + p} \quad (2)$$

The pressure is induced by oil film, which is generally starved [2]. Under this condition, the Hertzian dry contact pressure can be effectively used as an approximation of the pressure profile [41]:

$$p\left(x, z\right) = p_{Hz} \sqrt{1 - \left(\frac{x}{a}\right)^2 - \left(\frac{z}{b}\right)^2} \quad (3)$$

The term "thickener-rich layer" is commonly used to describe the deposited thickener layer on the contacting surfaces, as it contains a higher fraction of thickener compared to the fresh grease [9,40,42]. However, this thickener-rich layer still retains some base oil within its porous structure. Therefore, to predict the lateral side flow $\hat{q}_z$ of the thickener-rich layer, the aforementioned thin layer model is extended to a porous thin layer (PTL) model in this study.

In the PTL model, as shown in Fig. 2(c), the lateral side flow $\hat{q}_z$ includes two components: $\hat{q}_{th,z}$ the side flow of thickener-rich layer as a homogeneous mixture fluid of thickener and oil, and $\hat{q}_{oil,z}$ the side flow of the base oil itself within the porous structure of thickener-rich layer [27]. Therefore, $\hat{q}_z$ can be calculated as:

$$\hat{q}_z(z, t) = \hat{q}_{th,z} + \hat{q}_{oil,z} = \int_0^{L_{track}} \int_0^{\frac{h_{th}}{2}} (\rho w_{th}) dy dx + \int_0^{L_{track}} \int_0^{\frac{h_{th}}{2}} (\rho w_{oil}) dy dx \quad (4)$$

2.1.2. Side flow model of thickener-rich layer as a homogeneous fluid

The side flow in the contact area is induced by the contact pressure. A slip layer may occur between grease and the wall at low shear rates (lower than approximately $0.5\text{--}10^2 \text{ s}^{-1}$) [14]. In this work, the shear rate in the contact zone is approximately in the order of 10^5 s^{-1} . Therefore, the slip effect between thickener-rich layer and contact surface is

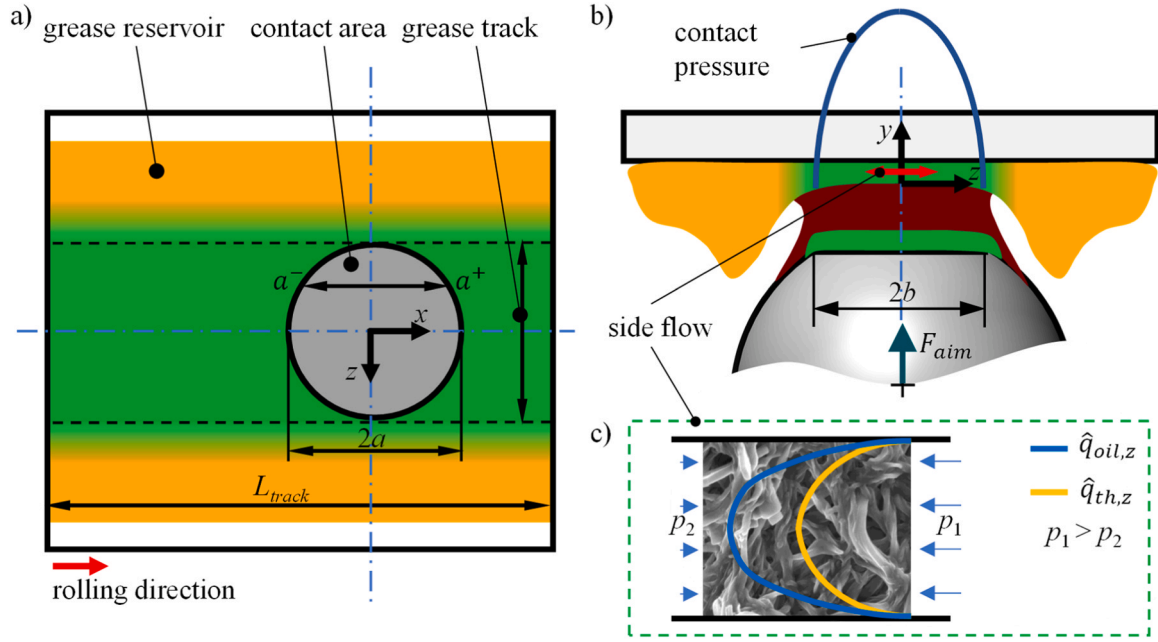


Fig. 2. Schematic representation of thickener-rich layer (a) Along rolling direction, (b) Along cross rolling direction and (c) Components of lateral side flow induced by the contact pressure.

neglected. As a homogeneous mixture fluid of thickener and oil, the flow velocity of the thickener-rich layer can be described by Eqs. (5)–(6):

$$u_{th}(y) = \frac{1}{2\eta_{th}(\dot{\gamma}, p)} \frac{\partial p}{\partial x} \left(y^2 - \frac{h_{th}}{2} y \right) + \frac{2(u_{disc} - u_{inter})}{h_{th}} y + u_{inter} \quad (5)$$

$$w_{th}(y) = \frac{1}{2\eta_{th}(\dot{\gamma}, p)} \frac{\partial p}{\partial z} \left(y^2 - \frac{h_{th}}{2} y \right) \quad (6)$$

where the u_{disc} and u_{inter} are the boundary conditions, which representing the velocity of the disc surface and the velocity of the interface between thickener-rich layer and oil film, respectively. In this work, u_{inter} can be approximated by the velocity of the ball u_{ball} .

The non-Newtonian viscosity of thickener-rich layer is calculated by the Carreau-Yasuda equation [28]:

$$\eta_{th,p0}(\dot{\gamma}) = (\eta_{th,\dot{\gamma}0,p0} - \eta_{oil,p0}) [1 + (\lambda \dot{\gamma})^{ay}]^{\frac{1}{ay}} + \eta_{oil,p0} \quad (7)$$

From the high-pressure rheology tests by Bair, the ratio of effective grease viscosity to base oil viscosity was fixed with respect to pressure [43]. Therefore, the pressure-viscosity behavior of thickener-rich layer can be formulated by Roelands equation with the same pressure-viscosity index as the base oil [28]:

$$\eta_{th}(\dot{\gamma}, p) = \eta_{th,p0} \bullet e^{\left\{ \left(\ln \eta_{oil,p0} + 9.67 \right) \bullet \left[\left(1 + \frac{p}{p_{ref}} \right)^z - 1 \right] \right\}} \quad (8)$$

where $\eta_{oil,p0}$ is the viscosity at ambient pressure, $p_{ref} = 1.96 \times 10^8$ Pa and z is the pressure-viscosity index:

$$z = \frac{p_{ref} \bullet \beta_{Barus}}{\ln \eta_{oil,p0} + 9.67} \quad (9)$$

where β_{Barus} can be measured using high-pressure viscometers. Meanwhile, according to Nogi et al. [28], Eq.(7) can be simplified for a rolling point contact because $\dot{\gamma} \gg 1/\lambda$:

$$\eta_{th,p0}(\dot{\gamma}) = \tau_{y,p0} \dot{\gamma}^{-1} \quad (10)$$

where $\tau_{y,p0}$ is the yield stress, and can be represented by a power function of the thickener fraction [28]. Using the tests results for the

lithium greases by Yeong et al. [44], it can be fitted as:

$$\tau_{y,p0} = \frac{M}{\lambda} f_{th}^N = 6.448 \times 10^6 \bullet f_{th}^{4.552} \quad (11)$$

The viscosity of the thickener is often some 1000 times higher than that of the oil [27]. In this case, the Poiseuille flow can be neglected and the velocity of the thickener-rich layer is linear [27]. Therefore, the shear rate of the thickener-rich layer in the contact area can be approximated by:

$$\dot{\gamma} = \sqrt{\left(\frac{du_{th}}{dy} \right)^2 + \left(\frac{dw_{th}}{dz} \right)^2} \approx \sqrt{\left(\frac{2(u_{disc} - u_{inter})}{h_{th}} \right)^2} = \frac{2u_{disc}\delta_{Hz}}{h_{th}R_{ball}} \quad (12)$$

Substitution of Eqs.(5) to (12) into Eq.(4) gives the side flow equation of thickener-rich layer as a homogeneous mixture of thickener and oil:

$$\hat{q}_{th,z}(z, t) = \frac{p_{Hz}^2}{96b^2} \bullet z \bullet \rho^3 h_{th}^3 \int_{a^-}^{a^+} \rho^{-2} \eta_{th}^{-1} p^{-1} dx \quad (13)$$

2.1.3. Side flow model of base oil within the porous thickener structure

Likewise in [27], the side flow of the base oil within the porous structure of thickener-rich layer is described by the Darcy's equation:

$$\frac{\partial p}{\partial z} + \frac{w_{oil}\eta_{oil}}{k} = \eta_{oil} \frac{\partial^2 w_{oil}}{\partial y^2} \quad (14)$$

Eq.(14) is an ordinary differential equation and can be solved by the standard method, as introduced in [27]. The flow velocity is obtained as:

$$w_{oil}(y) = \frac{1}{\eta_{oil}} \frac{\partial p}{\partial z} \bullet k \left[\frac{e^y / \sqrt{k} + e^{\left(\frac{h_{th}}{2} - y \right) / \sqrt{k}}}{e^{\left(\frac{h_{th}}{2} / \sqrt{k} \right) + 1}} - 1 \right] \quad (15)$$

where k is the permeability of the thickener. By assuming the thickener has a fiber structure, Baart et al. provided a model to calculate this parameter [4]:

$$k = \frac{2}{3} k_{\perp} + \frac{1}{3} k(\alpha) \quad (16)$$

$$k(\alpha) = \frac{k_{\perp} k_{//}}{k_{//} \sin^2 \alpha + k_{\perp} \cos^2 \alpha} \quad (17)$$

$$\begin{cases} k_{\perp} = \frac{16}{9\pi\sqrt{2}} \left(\sqrt{\frac{f_{th,max}}{f_{th}}} - 1 \right)^{5/2} r_{th}^2 k \\ k_{//} = \frac{8}{57} \frac{(1-f_{th})^3}{f_{th}^2} r_{th}^2 \end{cases} \quad (18)$$

In Eqs.(16)-(18), r_{th} is the mean fiber radius and α is mean fiber tilted angle, reflecting the variation of the fiber orientation during bleeding [4]. The evaluations of both parameters are done by the following formulas, respectively:

$$r_{th}^2 = \frac{r_{th,p0}^2}{\bar{\rho}} \quad (19)$$

$$\cos \alpha = \frac{f_{th}^{-1} - f_{th,max}^{-1}}{f_{th,0}^{-1} - f_{th,max}^{-1}} \quad (20)$$

where f_{th} is the thickener fraction of the thickener-rich layer. $f_{th,0}$ and $f_{th,max}$ are thickener fraction of the fresh grease and maximum thickener fraction, respectively. For the greases with a fiber structure, $f_{th,max}$ equals to $\pi/4$ [4]. In this study, pressure-viscosity relation of the base oil is evaluated by the Roelands equation:

$$\eta_{oil}(p) = \eta_{oil,p0} \bullet e^{\left\{ \left(\ln \eta_{oil,p0} + 9.67 \right) \bullet \left[\left(\frac{1 + \frac{p}{p_{ref}}}{1} \right)^z - 1 \right] \right\}} \quad (21)$$

Substituting of Eqs.(15) to (21) into Eq.(14), the side flow flux of the base oil within the porous structure of thickener-rich layer can be obtained:

$$\hat{q}_{oil,z}(z, t) = \frac{p_{Hz}^2}{b^2} \bullet z \bullet \int_{a^-}^{a^+} \left\{ \rho \eta_{oil}^{-1} p^{-1} \bullet k \left[\frac{h_{th}}{2} - \frac{2\sqrt{k} \left(e^{\frac{h_{th}}{2\sqrt{k}}} - 1 \right)}{e^{\frac{h_{th}}{2\sqrt{k}}} + 1} \right] \right\} dx \quad (22)$$

2.1.4. Film thickness evolution of thickener-rich layer

For the point contact in this study, the pressure is symmetric by the center line $z = 0$, see Eq.(3). In these cases, the film thickness of the thickener-rich layer following the center line $z = 0$ can be obtained by substituting of Eqs.(13) and (22) into Eq. (1), and simplified as:

$$\frac{\partial h_{th}}{\partial t}(x, z = 0, t) = -\frac{ap_{Hz}}{48L_{track}b^2} \frac{\bar{\rho}^2}{\eta_{th,p0}^3} h_{th}^3 C_{th} + \frac{2ap_{Hz}}{L_{track}b^2} \frac{1}{\rho \eta_{oil,p0}} C_{oil} \quad (23)$$

$$C_{th} = \int_{-1}^1 \bar{\rho}^{-2} \eta_{th}^{-1} [1 - (X)^2]^{-\frac{1}{2}} d(X) \quad (24)$$

$$C_{oil}(x, z = 0, t) = \int_{-1}^1 \bar{\rho}^{-1} \eta_{oil}^{-1} [1 - (X)^2]^{-\frac{1}{2}} k \left[\frac{2\sqrt{k} \left(e^{\frac{h_{th}}{2\sqrt{k}}} - 1 \right)}{e^{\frac{h_{th}}{2\sqrt{k}}} + 1} - \frac{h_{th}}{2} \right] d(X) \quad (25)$$

where X is dimensionless x coordinate, $\bar{\rho}$ is dimensionless density and $\bar{\eta}$ is dimensionless viscosity.

2.2. Porous multiphase bleeding model

2.2.1. Flow model of oil infiltration between two contacts

The stationary reservoirs of grease deliver oil through the thickener-rich layer to the contact because of a so-called infiltration effect [36], see Fig. 1. In this study, a porous multiphase bleeding model (PMB) is used to simulate this oil infiltration process. This PMB model is based on an open-source toolbox for multiphase flow in porous medium proposed by Horgue et al. [45]. In this model, the oil saturation in the void space of porous medium is first defined as:

$$S_{oil} = \frac{V_{oil}}{V_{void}} = \frac{V_{oil}}{\varepsilon \bullet V_{cell}} = 1 - S_{air} \quad (26)$$

where V_{oil} and V_{void} are the volume of the oil and void space in each mesh cell, V_{cell} is the volume of this mesh cell, ε is porosity of the grease in between two contacts. In the porous structure of the thickener-rich layer, the oil infiltration, represented by the variation of S_{oil} , can be determined by the mass conservation equation and the generalized Darcy's model [45]:

$$\varepsilon \frac{\partial S_i}{\partial t} + \nabla \bullet \mathbf{U}_i = 0 \quad (27)$$

$$\mathbf{U}_i = -\frac{K_i(S_{oil})}{\mu_i} \bullet (\nabla p_i - \rho_i \mathbf{g}) \quad (28)$$

In Eqs.(24) and (25), the i is the representation of air or oil phase. K_i is the apparent permeability and the ∇p_i is the pressure gradient driving the flow of air or oil. The capillary pressure p_c is defined as the pressure difference between the air phase p_{air} and oil phase p_{oil} [45,46]:

$$p_c(S_{oil}) = p_{air} - p_{oil} \quad (29)$$

In this study, the Brooks and Corey model [47] and linear model [45] are used to determine K_i and p_c , respectively:

$$\begin{cases} K_{air} = k \bullet (1 - S_{oil})^m \\ K_{oil} = k \bullet (S_{oil})^m \end{cases} \quad (30)$$

$$p_c(S_{oil}) = (1 - S_{oil})p_{c,max} \quad (31)$$

where m is a constant power coefficient and $p_{c,max}$ is the maximal capillary pressure. k is the permeability of the thickener structure, which can be calculated by Eqs.(16)-(18).

The final equations are obtained by substitution of Eqs.(26), (28)-(31) into Eq.(27) and reformulated into a pressure-saturation system [45]:

$$\nabla \bullet \left(-\frac{K_{air}}{\mu_{air}} (\nabla p_{air} - \rho_{air} \mathbf{g}) \right) + \nabla \bullet \left(-\frac{K_{oil}}{\mu_{oil}} (\nabla p_{oil} - \rho_{oil} \mathbf{g} - \nabla p_c) \right) = 0 \quad (32)$$

$$\varepsilon \frac{\partial S_{oil}}{\partial t} + \nabla \bullet \mathbf{U}_{oil} = 0 \quad (33)$$

The p_{air} and S_{oil} are the two coupling variables in Eqs.(32) and (33), which can be solved by the *Implicit Pressure Explicit Saturation* (IMPES) approach in the open source CFD package *OpenFOAM* [45]. Pressure and velocity coupled equations are solved by the PCG matrix solver with DIC smoother, and the tolerances were set as 10^{-12} [48]. To increase the simulation efficiency, the *runTimeModifiable* is set as *yes* and the *CFL* condition is used to limit the time step [45]. Together with a discretization on the finite volume mesh and properly defined boundary conditions, the transient variation of the S_{oil} over time can be calculated. Therefore, the oil infiltration through the thickener-rich layer can be simulated.

2.2.2. Model setup

For a ball-on-disc contact (see Fig. 3(a)), the ball forms a grooved track into the deposited grease on the raceway after the churning phase [22], and subsequently greases tend to settle at the side of the contact [36,49]. Therefore, the stationary grease around the contact in bleeding phase can be meshed as shown in Fig. 3(b). The grease surface, which is at the bottom of the model, is assumed as a fixed cylindrical surface. The opposite side of the track is a flat disc surface. During the overrolling, the thickener-rich layer is gradually formed in the grease track zone along rolling direction (x direction in Fig. 3(b)). As previously mentioned, the permeability and the thickener fraction of this layer are varying over time.

To increase the simulation efficiency, only half of the grease

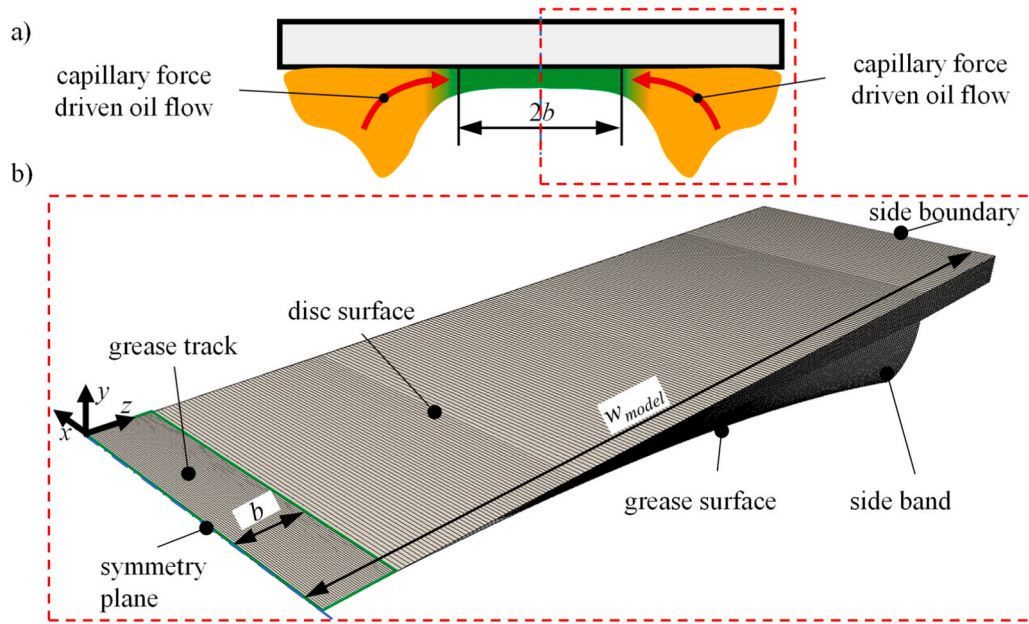


Fig. 3. Setup of PMB model (a) schematic representation of grease distribution (b) mesh and boundaries for the PMB model.

reservoir is modeled in this study and a *symmetryPlane* boundary condition is used at $z = 0$. The size of the simulation domain is determined by the method described in the appendix. Moreover, the simulations in this study are carried out at 0.2 m/s, therefore, the centrifugal force can be neglected. To ensure that the oil flow inside the thickener structure is

only driven by capillary pressure, a pressure boundary condition of 0 Pa is applied to all surfaces.

To ensure that the oil infiltration results are independent on the size of the simulation domain, the model width w_{model} is checked for various values. Meanwhile, the mesh independence study is carried out to make

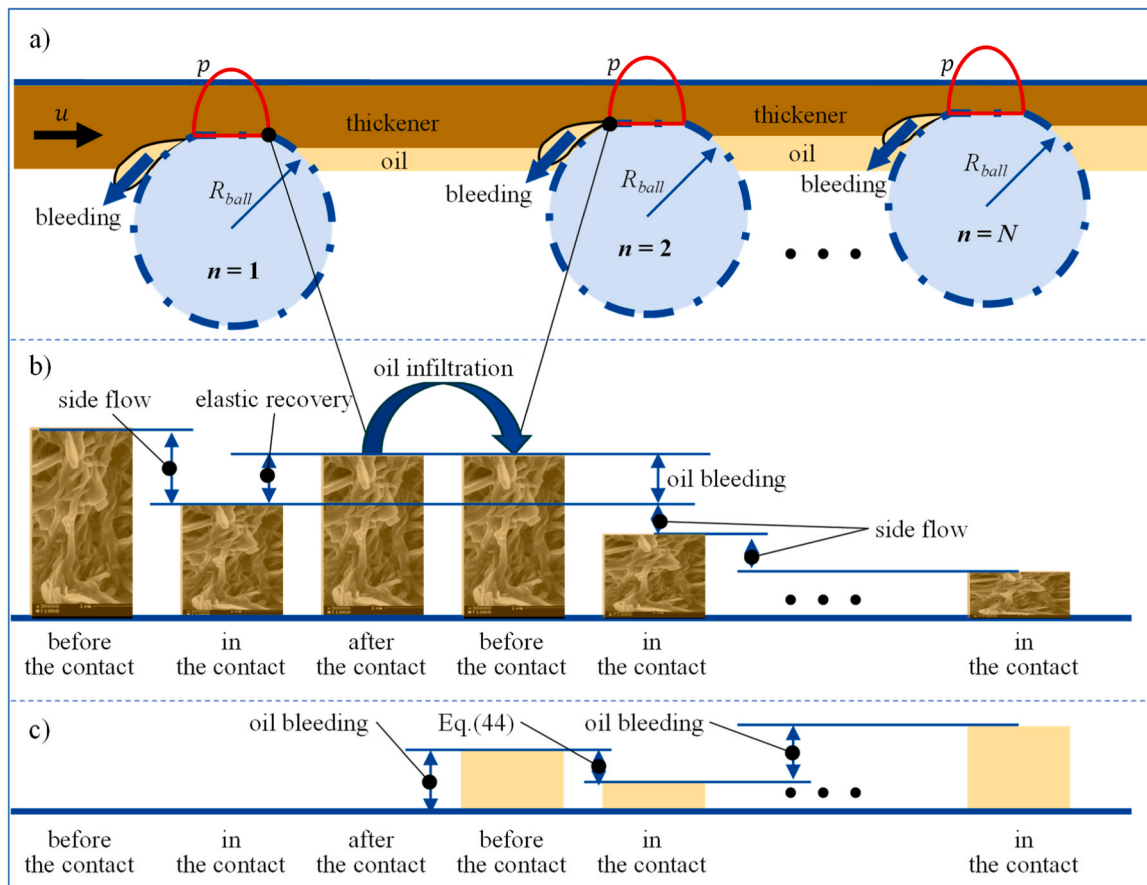


Fig. 4. Schematic representation of lubrication mechanism after churning phase following the center line $z = 0$ (a) Formation of two-component grease film, (b) Thickness evolution of the thickener-rich layer and (c) Thickness evolution of the oil film.

sure the simulation results by the PMB model is not affected by the mesh cell number [22,50]. The results of the model size and mesh independence studies are also shown in the appendix.

3. Solution procedure

When the bulk grease has been displaced from the running track, and the lubricating film thickness is composed of the deposited thickener-rich layer and a bled oil film, the bleeding phase begins [14]. The objective of this work is to predict the grease film thickness in the bleeding phase. As the first attempt, the ball-on-disc rolling contact is modeled. The surface roughness in this condition is relatively low, therefore, its effect is neglected in this work.

The solution procedure is organized according to the grease lubrication mechanisms in the bleeding phase, as demonstrated in the Fig. 4. At first, there is only thickener-rich layer in front of the contact, which is similar to the fresh grease, as shown in Fig. 4(a). Then, this layer is compressed within the contact and two types of the side flows occur due to the contact pressure. As a result, the thickness of the thickener-rich layer decays. These side flows in the contact zone are modeled by the PTL model, as introduced in Section 2.1.

During the simulations, the load is balanced by the Hertzian dry contact pressure, which is expressed by Eq.3. In the contact area, the central film thickness $h_{th,c}$ converges to [41]:

$$h_{th} = \frac{\rho_{Hz} h_{th,c}}{\rho} \quad (34)$$

Substitution of Eq. (34) into Eq. (23) gives the central film thickness evolution over time:

$$\frac{\partial h_{th,c}}{\partial t}(t) = -\frac{ap_{Hz}}{48L_{track}b^2} \frac{\bar{\rho}_{Hz}^2}{\eta_{th,p0}} h_{th,c}^3 C_{th} + \frac{2ap_{Hz}}{L_{track}b^2} \frac{1}{\bar{\rho}_{Hz}\eta_{oil,p0}} C_{oil} \quad (35)$$

$$C_{th} = \int_{-1}^1 \bar{\rho}^{-2} \bar{\eta}_{th}^{-1} [1 - (X)^2]^{-\frac{1}{2}} d(X) \quad (36)$$

$$C_{oil}(t) = \int_{-1}^1 \bar{\rho}^{-1} \bar{\eta}_{oil}^{-1} [1 - (X)^2]^{-\frac{1}{2}} k \left[\frac{2\sqrt{k} \left(e^{\frac{\rho_{Hz} h_{th,c}}{2\rho\sqrt{k}}} - 1 \right)}{e^{\frac{\rho_{Hz} h_{th,c}}{2\rho\sqrt{k}}} + 1} - \frac{\rho_{Hz} h_{th,c}}{2\rho} \right] d(X) \quad (37)$$

Meanwhile, the film thickness decay due to the side flow of the thickener-rich layer can be calculated by Eq. (38):

$$\frac{\Delta h_{th,c}^{*,t}}{\Delta t} = -\frac{ap_{Hz}\delta_{Hz}}{24b^2} \frac{\bar{\rho}_{Hz}^2}{l_t} \frac{u_{disc}}{R_{ball}} \bullet \frac{h_{th,c}^{*,t}}{\tau_y^0} \bullet C_{th} \quad (38)$$

Therefore, the thickener fraction of next time step $f_{th}^{t+\Delta t}$ in the contact center can be determined as:

$$f_{th}^{t+\Delta t} = \frac{(h'_{th} - \Delta h_{th,c}^{*,t})f_{th}^t}{h_{th}^{t+\Delta t}} \quad (39)$$

Above equations are solved for the evolution of the thickener-rich layer in terms of time t . Then, the results are interpolated towards to a finite number of disc revolution n to simulate the oil infiltration happening between two contacts, see Fig. 4(b). In this study, n is set to 1 to start the simulation and its maximum value is equal with the experimental conditions.

As presented by Fig. 4(b), after the contact, the thickener-rich layer can spring back to a higher level than thickness in the contact zone, due to the compressibility of grease and the elastic recovery of thickener [51]. This elastic recoverability is evaluated by the thin elastic liner model proposed by Hamel et al. [52,53]. Therefore, the thickness of the thickener-rich layer between two contacts at disc revolution number n can be determined as:

$$\tilde{h}_{th,c}^n = \max \left[\frac{h_{th}^n \bar{\rho}}{2}, \frac{h_{th,c}^n}{2(1 - ke_{th} p_{Hz})} \right] \quad (40)$$

where ke_{th} is a parameter related with the elasticity of thickener structure. This property is difficult to measure [9]. Therefore, as a first attempt, ke_{th} is approximated by assuming that the thickener-rich layer can recover to the initial film thickness, which means the $\tilde{h}_{th,c}^1 = \tilde{h}_{th,c}^0$.

Then, according to the mass conservation of thickener, the thickener fraction between two contacts can be calculated by:

$$\tilde{f}_{th}^n = \frac{\bar{\rho} h_{th,c}^n f_{th}^n}{2\tilde{h}_{th,c}^n} \quad (41)$$

When the elastic recovery is higher than film compression due to the compressibility, a reduction of the oil saturation after this contact S_{oil} on the track can be resulted as:

$$S_{oil}^n = \frac{V_{oil}}{V_{void}} = \frac{h_{th}^n \bar{\rho} (1 - f_{th}^n)}{2\tilde{h}_{th,c}^n (1 - \tilde{f}_{th}^n)} \quad (42)$$

Therefore, the capillary pressure is generated in the thickener-rich layer between two contacts and the grease reservoirs at the contact sides can deliver the base oil to the track center, as shown in Fig. 4(b). This infiltration process is simulated with the PMB model, which has been introduced in Section 2.2. Eqs.(40)-(42) are used as the input parameters. After the simulation of infiltration, the oil saturation before the next contact $\tilde{S}_{oil}^n$ can be determined.

With the subsequent overrolling, the oil, that has infiltrated from the grease reservoirs is pressed out in front of the contact, see Fig. 4(b). The bled oil volume V_{oil} at disc revolution number n can be calculated as:

$$\frac{V_{oil}^n}{L_{track}} = \sum_1^n \frac{\Delta V_{oil}^n}{L_{track}} = \sum_1^n \int_{-a}^a \tilde{h}_{th,c}^n (1 - \tilde{f}_{th}^n) \bullet (\tilde{S}_{oil}^n - S_{oil}^n) dx \quad (43)$$

where V_{oil}/L_{track} is the oil volume factor, describing the available oil volume related to the track length [21]. When the bled oil is available in the vicinity of contact, the oil film is formed in the contact, see Fig. 4(c). Its thickness is calculated by starvation degree method proposed by Cann et al. [33].

$$\frac{h_{s,c}^n}{h_{ff,c}^n} = \left(\frac{A \bullet h_{oil}^n}{Ca \bullet a} \right)^B \quad (44)$$

where A and B are constant values, but can be different for various conditions [19,34]. For a single EHL contact, A is observed between 1.5 and 2 [33], B can be 1.67 [33] or 1.31 [19]. h_{oil}^n is the oil height of the side band in the vicinity of the track at disc revolution number n . Considering the changes of contact angle by the thickener-rich layer [21, 22,24,54], h_{oil}^n is calculated by assuming the side band has a partial circular profile[19]:

$$R_{sb} = \sqrt{\frac{2V_{oil}^n/L_{track}}{2\theta - \sin(2\theta)}} \quad (45)$$

$$h_{oil}^n = R_{sb}[1 - \cos(\theta)] \quad (46)$$

where θ is the contact angle between the bled oil and the corresponding thickener-rich layer.

Simultaneously with the oil film formation, the side flows of thickener-rich layer $\hat{q}_{th,z}$ and the oil within this layer $\hat{q}_{oil,z}$ occur repeatedly. Therefore, the above process will be repeated within the next disc revolution in order to predict the transient film thickness evolution. The solution procedure of above equations is summarized as Fig. 5. Eventually, the film thickness evolution of the grease film can be predicted by treating it as a superposition of the thickener-rich layer (by Eqs.(35)-(37)) and the oil film (by Eq.(44)).

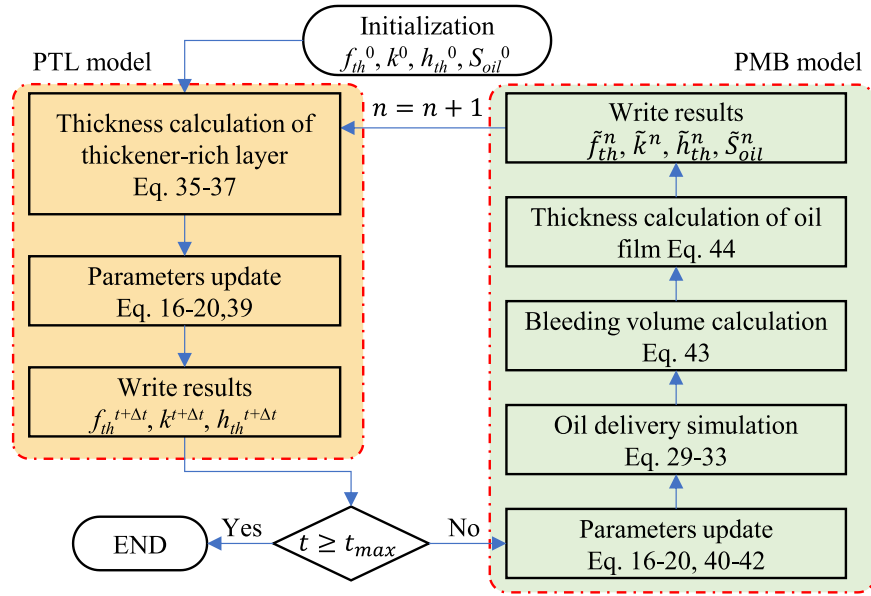


Fig. 5. Simulation flowchart (PTL model is short for Porous Thin Layer model and PMB model is short for Porous Multiphase Bleeding model. $t_{\max}$ is equal with the experimental conditions, which is about 100 minutes).

4. Experimental methods and materials

In a rolling contact, the grease film thickness evolution can be measured using the ball-on-disc tribometer. Fig. 6 illustrates the experimental setup used in this study. To determine the central film thickness, a spectrometer records the interference two light beams reflected by the bottom of the glass disc and the surface of the steel ball, respectively. Additionally, an RGB camera was utilized to capture images of thickener particles in the contact zone. Further details about this tribometer can be found in [11,23,55].

Before conducting each test, a grease distributor was used to ensure uniform distribution of grease on the disc. Approximately 0.1 mm of grease was applied for each test, as described in [11]. The tests were conducted with the parameters listed in Table 1 for 5000 disc revolutions, which corresponds to approximately 100 minutes. The grease film thickness tests were repeated three times for each type grease with new specimens for ball and disc track positions. At the end of each test, the motor was stopped and the thickness of the thickener-rich layer was measured at zero speed and under operating load at six angularly distributed positions. L_{track} equals to the cycle length of the grease track on the disc in this work, which is listed in Table 1.

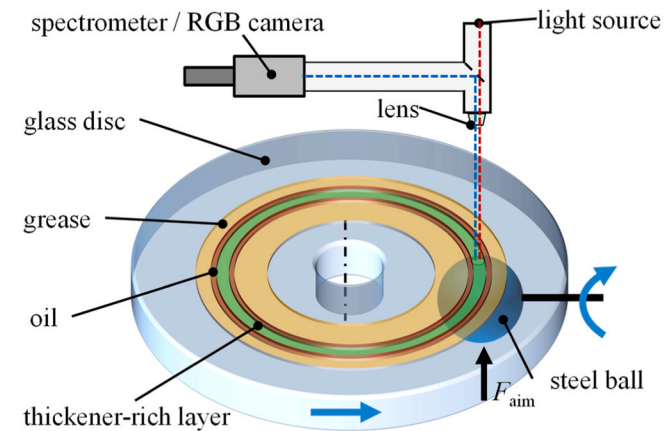


Fig. 6. Schematic setup for the measurement of a grease-lubricated ball-on-disc contact.

Table 1

Parameters of the lubricant and operation condition.

Lubricant Parameter	Value	Operation Parameter	Value
Base oil	ρ_{oil}^0	F_{aim}	47 N
PAO	ρ_{oil}	R_{ball}	9.525 mm
	β_{Barus}	E_{ball}	207 GPa
	η_{air}	v_{ball}	0.29
	ρ_{air}	$R_{q,ball}$	6.1 nm[56]
Thickener	f_{th}	E_{disc}	75 GPa
Li-C	r_{th}	v_{disc}	0.22
	θ	$R_{q,disc}$	0.8 nm[56]
	NLGI	u_{disc}	0.2 m/s
Thickener	f_{th}	SRR	0
Li-S	r_{th}	L_{track}	0.242 m
	θ	$h_{oil,ff}$	43.9 nm
	NLGI	h_{Λ}	18.5 nm

Permeability is a crucial parameter for quantifying oil bleeding behavior. However, the complexity of thickener structures in different greases have resulted in the lack of a standardized method for calculating this parameter [5]. In this study, Baart's model, which is one of the widely used permeability models [25,35,57], was employed, as introduced in Section 2. In this model, it is assumed the thickener has a fibrous structure with a relatively high-volume fraction [4]. Therefore, two lithium greases, which have fibrous structures as shown in Fig. 7, are chosen for validation. The greases have NLGI classes 2 and 3 respectively, which are widely used for the bearing lubrication [14]. In this study, Li-C represents lithium complex thickener and Li-S represents lithium simple thickener. The fiber radii are provided in Table 1.

Once the thickener-rich layer forms on the track, the contact angle between base oil and the contacting surface undergoes changes, which significantly affects the formation of the oil film, as discussed in [21,22,24]. The contact angles between the PAO base oil and both thickener-rich layers are shown in Fig. 8. These tests were conducted three times and the mean values were listed in Table 1. The obtained contact angles align with the findings in [24]. Notably, the deposition of thickener has a substantial impact on the contact angle. Specifically, the contact angle between PAO oil and Li-C thickener layer exhibits a significant higher value when compared to the same base oil and Li-S thickener layer. Therefore, it can be expected that the oil film on Li-C

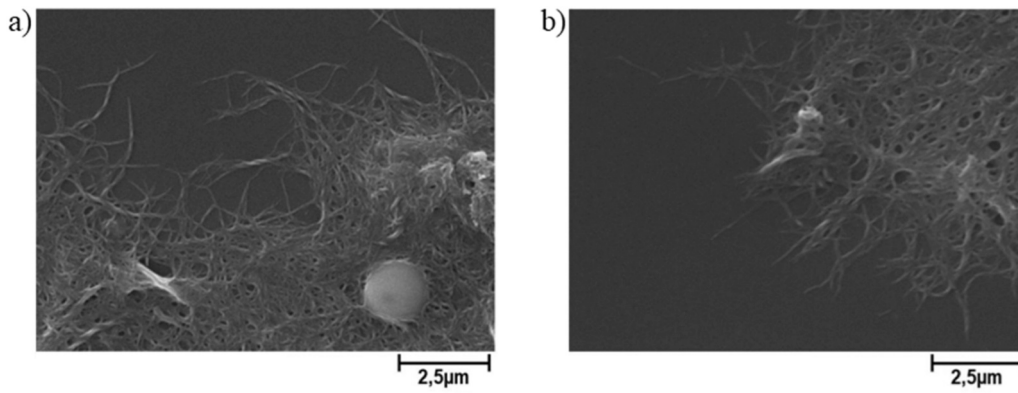


Fig. 7. Thickener structure of two greases (a) PAO-Li-C (b) PAO-Li-S.

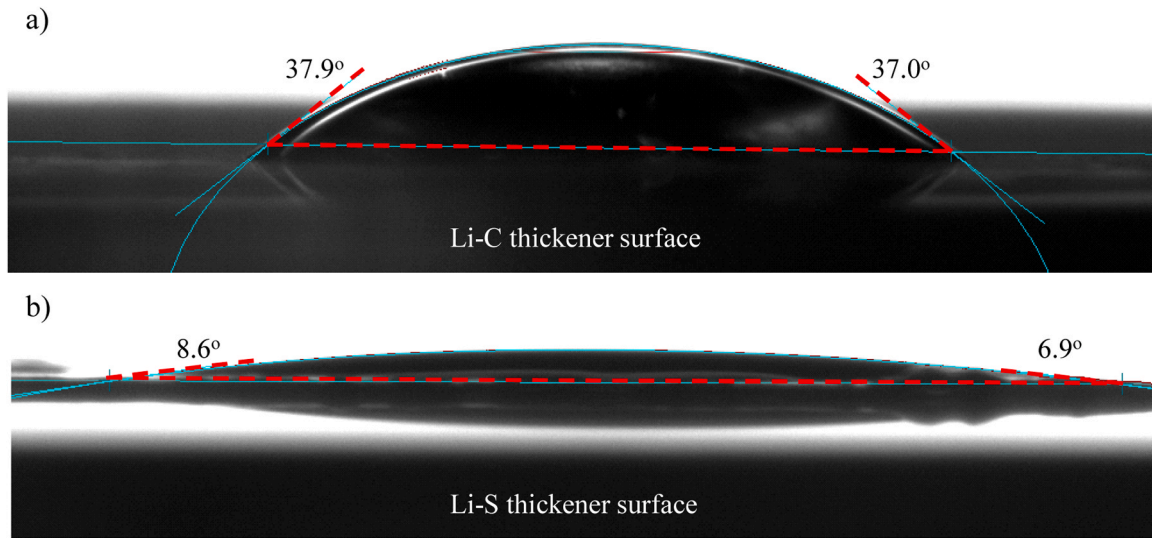


Fig. 8. Steady contact angle of the base oil on different thickener layers (a) Li-C thickener layer (b) Li-S thickener layer.

thickener-rich layer shows a higher film thickness with same bleed oil volume, according to the results in [21,22].

5. Results and discussion

In the following, the film thickness evolution predicted by the proposed models is analysed to clarify the film evolution mechanisms in a grease-lubricated rolling contact. Furthermore, as a key factor influencing the grease lubrication, the effects of thickener on the film formation evolution are also investigated. For validation, the numerical results are compared with experimental data.

5.1. Evolution mechanisms of grease film

The film thickness evolution during overrolling using PAO-Li-C as a lubricant is shown in Fig. 9. For a better understanding of lubrication condition, the oil film thickness under fully flooded condition $h_{oil,ff}$ is calculated by the fully flooded Hamrock-Dowson equations using the base oil parameters [58]. The critical film thickness for mixed-lubrication condition h_A is also calculated by the 3 Λ -ratio method [56]. Due to the uncertainty of the constant values A and B in Cann's equation Eq. 44 [19,33,34], the possible oil film thickness h_{oil} is depicted with a max-min span. At the beginning of Fig. 9, the grease film thickness h_{total} mainly consists of the thickener-rich layer h_{th} and decays over time. After a certain period, an increasing oil film thickness h_{oil} can be observed, leading to a recovery of h_{total} . Therefore, the grease film

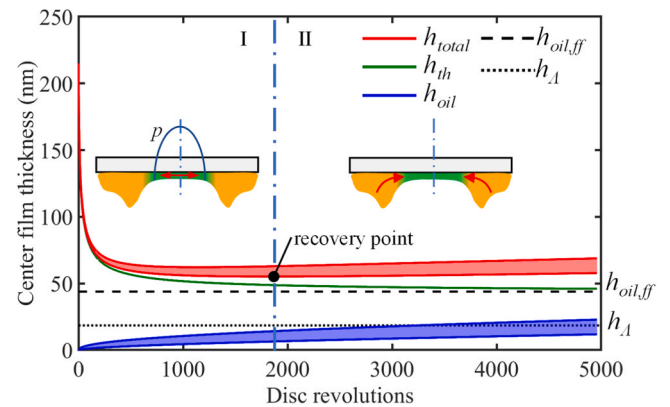


Fig. 9. Predicted film thickness evolution for PAO-Li-C (the predicted grease film thickness h_{total} is the superposition of the thickener-rich layer thickness h_{th} and oil film thickness h_{oil} ; when calculating oil film thickness, $A = 1.5$ and $B = 1.67$ for lower limit, $A = 2.0$ and $B = 1.31$ for upper limit).

thickness evolution during the bleeding phase can be separated into two zones by the recovery point occurring at around 1880 revolutions, which corresponds to 448.9 m or 37.6 min for the given set of parameters.

5.1.1. Evolution of thickener-rich layer

To elucidate the transient formation mechanisms of the grease film, the deposition process of the thickener-rich layer is firstly investigated. In the contact zone, lubricants are squeezed and flow laterally to sides due to the contact pressure gradient [41], see the flow along z -direction in Fig. 2. This side flow, denoted as $\hat{q}_z$ is a combination of two components: the side flow of the thickener-rich layer $\hat{q}_{th,z}$, and the side flow of base oil within the thickener-rich layer $\hat{q}_{oil,z}$, as shown in Fig. 10(a). As a result of mass conservation, the h_{th} decays in Fig. 9 zone I.

Meanwhile, as shown by Nogi et al. [28] and Tichy et al. [27], the side flow of thickener-rich layer $\hat{q}_{th,z}$ depends on its viscosity. In contrast, the side flow of the base oil $\hat{q}_{oil,z}$ is predominantly controlled by the permeability of the thickener-rich layer. At the beginning of bleeding phase, the thickener-rich layer has a relatively high permeability, and a significantly higher viscosity than the base oil, see Fig. 10(b). Consequently, the side flow gradient of oil, denoted as $\frac{\partial \hat{q}_{oil,z}}{\partial z}$ in Fig. 10(a) dominates over the gradient of $\hat{q}_{th,z}$. This aligns with findings from previous studies [27,28], which show that $\hat{q}_{oil,z}$ is much higher than $\hat{q}_{th,z}$. According to Eq. (1), a larger side flow gradient at contact center leads to a rapid decay of the center film thickness over time. Therefore, the thickness decay of h_{th} can mainly be attributed to the side flow of the oil within the thickener-rich layer under contact pressure.

Furthermore, it can be observed from Fig. 10(a) that the thickener fraction f_{th} in the contact zone increases with overrollings, as a result of the oil side flow $\hat{q}_{oil,z}$. Subsequently, as shown in Fig. 10(b), the viscosity of the thickener-rich layer increases over time and its permeability drops dramatically. Both factors, the viscosity increase and permeability decrease, lead to a decline of the oil side flow $\hat{q}_{oil,z}$, as observed by the reduced side flow gradient in Fig. 10(a). Therefore, the value of h_{th} gradually stabilizes after 4000 overrollings as shown in Fig. 9 zone II, and the thickener-rich layer behaves like a solid layer [9,24].

Turning our attention towards the thickener-rich layer properties after the contact, where the Hertzian compression is lifted and an elastic recovery of the layer height occurs [51]. Before the first overrolling, the thickener-rich layer is similar to the fresh grease, thereby its oil saturation is 1. After the first overrolling, the oil saturation S_{oil}^1 is 0.748, as shown in Fig. 11(a). This indicates the volume of the void space in the thickener-rich layer increases after the elastic recovery. Consequently, the thickener fraction after the contact $\tilde{f}_{th}$ is reduced, when compared to that in the contact zone f_{th} at same revolution number, see Fig. 10(a). Consequently, as shown in Fig. 11(b), the permeability after the contact

increases. Simultaneously, the void space in thickener-rich layer is occupied by ambient medium, viz., air from the surroundings [15] and the capillary pressure is generated between the air and oil phase [38]. At the beginning of disc revolution, the capillary pressure in the track center p_c is 6.661 kPa, in agreement with the experimental results of approximately 6 kPa by Zhang et al. [37]. As a result, the oil within the porous structure will be redistributed by the capillary pressure between two successive overrollings [59].

5.1.2. Evolution of oil film

In the bleeding phase, the oil film thickness is highly dependent on the amount of oil bleeding from the reservoirs [13,16]. The bleeding process can be induced by capillary pressure, which is shown in Fig. 12. It is observed that capillary pressure is generated in the thickener-rich layer on the track, while the capillary pressure far from the track center is nearly 0. Consequently, the generated capillary pressure drives the oil to flow from the track side to the track center through the thickener-rich layer.

The flow speed of oil within the thickener-rich layer between two overrollings at various disc revolutions are displayed by Fig. 13. It should be noted that the results are mirrored on the symmetry plane to show the entire flow field around the track. As shown by positive and negative $u_{z,oil}$ values with respect to the z -axis, the oil gradually flows from the grease reservoirs located at the sides to the track center. This flow is referred to as an oil infiltration through the thickener-rich layer.

Fig. 14(a) shows the oil saturation after the first contact and before the next one. The oil saturation after the first contact S_{oil}^1 is 0.748 and is uniform in the track. The maximum oil infiltration speed can be observed at the track side (see Fig. 13), where the largest gradient of oil saturation is found. As a result, oil saturation before the next contact S_{oil}^2 increases at the track side, when compared to the distribution of S_{oil}^1 . In order to show this increase clearly, the oil saturation along z -axis at the track center is shown in Fig. 14(b). The black circle indicates the position, where the oil infiltration can reach before the next overrolling. At the beginning of disc revolutions ($n = 1$), the track from $z/b = 0.08$ to $z/b = 1.00$ is wetted by the oil infiltration.

Moreover, the variation of the oil infiltration speed during the operation can also be observed from Fig. 13 and Fig. 14(b). When $n = 1$, the oil infiltration speed is relatively fast (around $2.3 \mu\text{m/s}$ at the track side). Therefore, more than 90% of the track center is wetted. With the subsequent overrolling, the infiltration speed becomes slower (around $0.33 \mu\text{m/s}$ at $n = 500$, around $0.21 \mu\text{m/s}$ at $n = 3000$, see Fig. 13). The

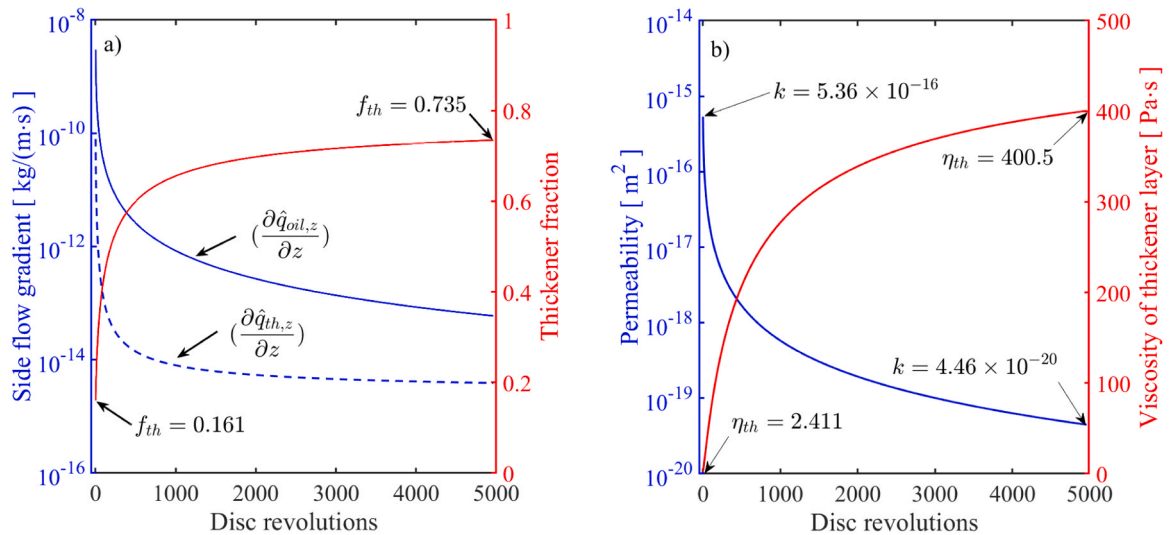


Fig. 10. Evolutions of the porous behaviors of the thickener-rich layer at the contact center (a) Side flow gradient and thickener fraction (b) Permeability and viscosity.

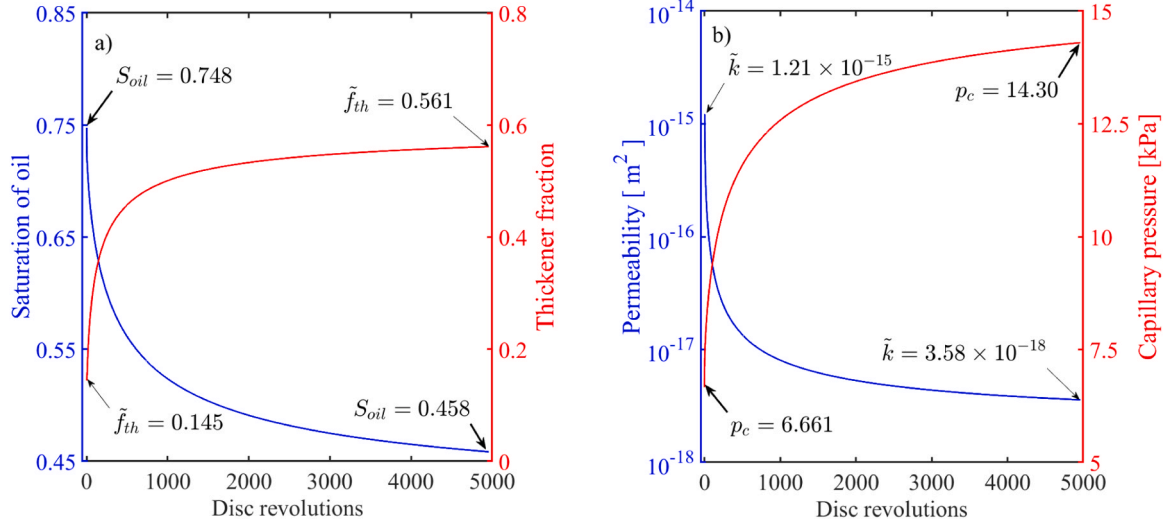


Fig. 11. Evolutions of the porous behaviors of the thickener-rich layer after the contact (a) Oil saturation and thickener fraction (b) Permeability and capillary pressure.

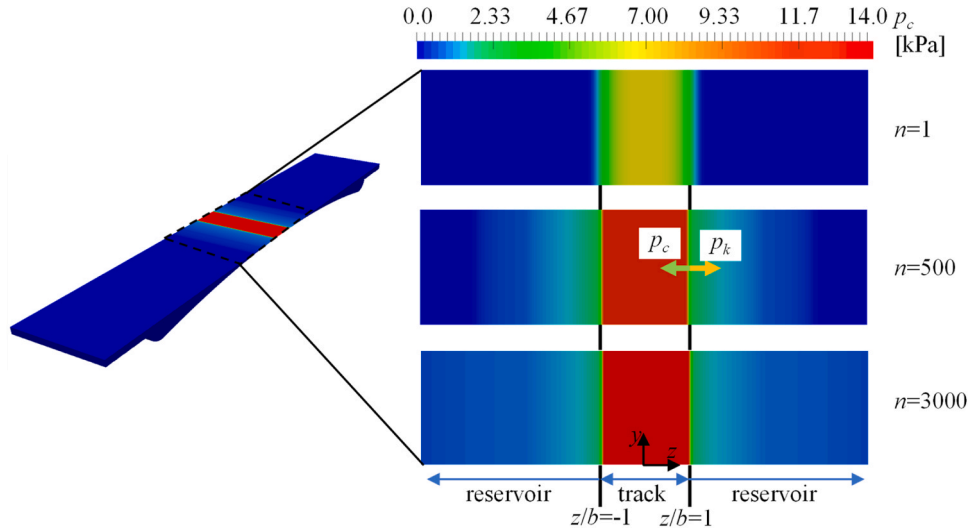


Fig. 12. Capillary pressure of the oil within the thickener-rich layer for different disc revolutions (the track is marked with black lines).

increase of oil saturation in the track center after infiltration becomes less and less, as indicated by the position of the black circle when $n = 500$ and $n = 3000$ in Fig. 14(b).

This variation can be explained by the balance between the driven pressure due to the capillary effects p_c and the resistance due to the porous effects p_k , as indicated in Fig. 12. With the increasing of revolutions, a significant drop of permeability and a rise in the capillary pressure can be observed simultaneously in Fig. 11(b). However, the oil side flow caused by the Hertzian contact pressure becomes degressive see Fig. 10(a). Consequently, the driven pressure p_c stabilizes eventually in Fig. 12, and the resistance p_k becomes the dominant factor on the oil bleeding rate.

The oil, that has infiltrated between two disc revolutions, is pressed out by the subsequent overrolling and becomes bled oil. Fig. 15 illustrates the oil bleeding rate and the corresponding bled oil volume during operation. As mentioned earlier, the oil infiltration speed gradually decreases due to the decrease of permeability. Consequently, the oil bleeding rate $\Delta V_{oil}/L_{track}$ also slows down and eventually stabilizes with increasing disc revolutions. By summing the oil bleeding amount for each disc revolution, the total bled oil amount V_{oil}/L_{track} can be determined, which has a similar power function pattern as the experimental

observations by Zhang et al. [37]. Consequently, the oil film gradually forms and the grease film thickness recovers, as shown in the Fig. 9.

Hence, the relevant mechanisms appear to be captured by this model. Specifically, the unidentified driven forces of oil bleeding from the stationary grease reservoirs mentioned in [15] can be attributed to the capillary pressure generated within the thickener-rich layer. Additionally, the physical nature of the infiltration effect, as discussed in [36] is the capillary pressure driven oil flow within a porous medium. For comparison, the bleeding behavior of PAO-Li-S are shown in Fig. 15 as well. The difference in bleeding behavior for the two greases can be explained by the variations of thickener fraction and fiber radius, which will be discussed with more details in the next section.

5.2. Effects of thickener on film thickness evolution

The effects of the thickener fraction and fiber radius on film thickness evolution are shown in Fig. 16. During the initial churning phase, the thickener is entrained into the contact. The relative increase in the film thickness, resulting from this entrainment of the thickener, is found to be proportional to the ratio of thickener fraction to the size of the fibers [60]. Consequently, a thinner thickener-rich layer is formed when the

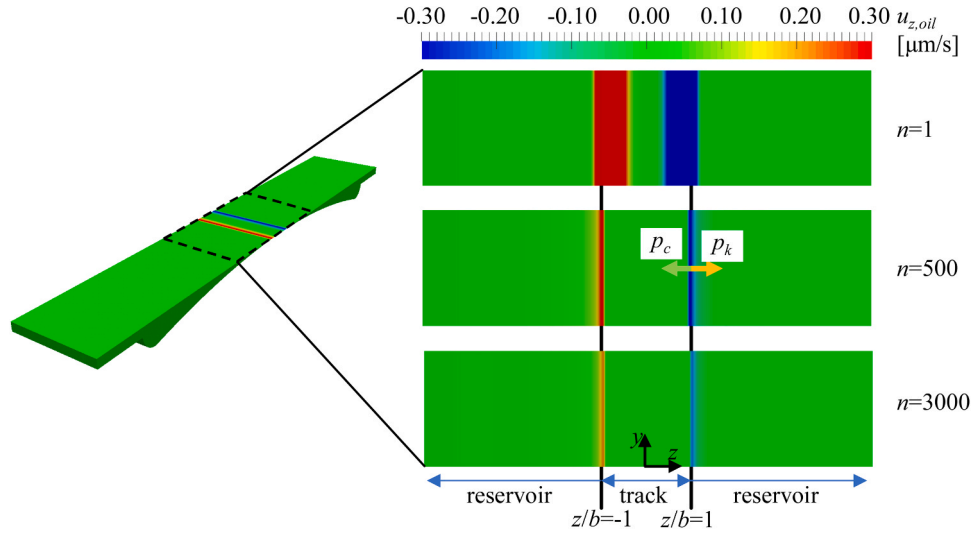


Fig. 13. Flow speed of the oil within the thickener-rich layer for different disc revolutions (the track is marked with black lines).

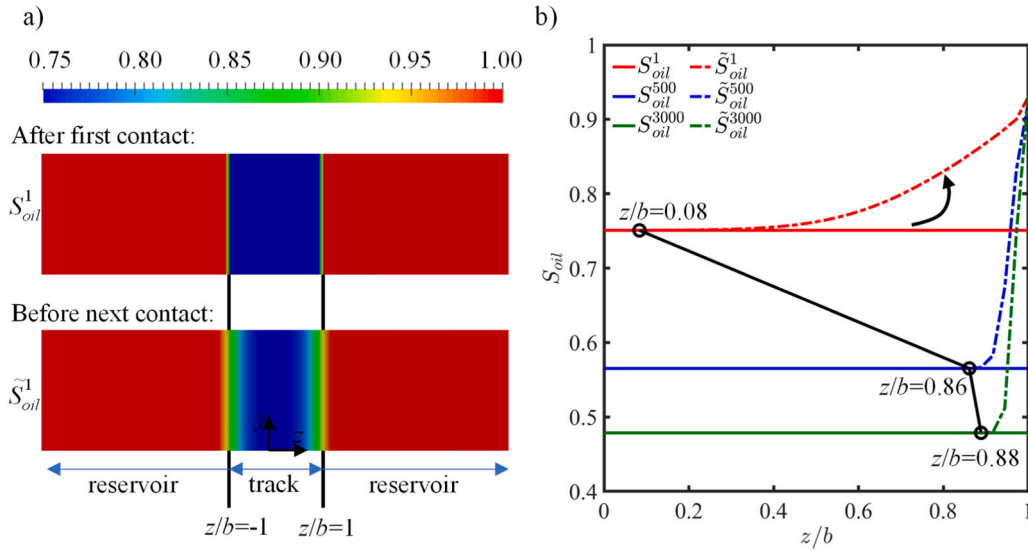


Fig. 14. Evolution of the oil saturation within the thickener-rich layer (a) Distribution of the oil saturation after contact of the first disc revolution S_{oil}^1 and before the contact of the second disc revolution $\tilde{S}_{oil}^1$ (the track is marked with black lines), (b) Oil saturation at track center for different disc revolutions.

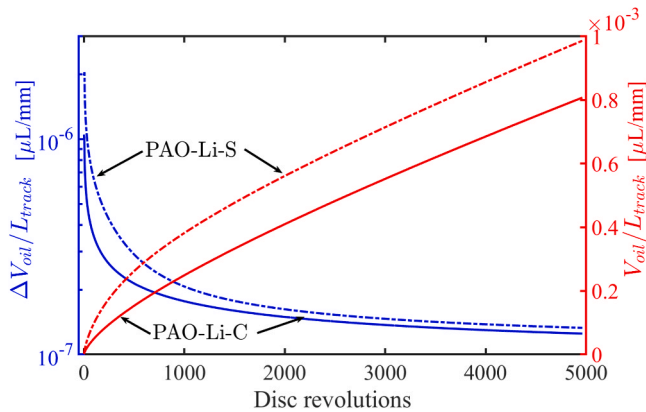


Fig. 15. Oil bleeding rate and bleeding volume during overrolling.

fresh grease has a lower thickener fraction or has a coarser fibers radius. Before the recovery points, which are indicated by black circles in Fig. 16, the thickener-rich layer has a dominant contribution to the grease film and is anticipated to have a sufficient thickness in order to effectively separate the contacting surfaces. However, at the recovery point, the grease film thickness decreases to 25.5 nm ($f_{th}=8.50\%$ in Figs. 16a), 24.8 nm ($r_{th}=45$ nm in Figs. 16b) and 23.7 nm ($r_{th}=65$ nm in Fig. 16b), which is only about 7 nm higher than the critical film thickness h_A for the ball-on-disc tribometer. Considering the higher surface roughness of the bearing surfaces [56], this situation poses an increased risk of wear under initial operating conditions.

Therefore, a film thickness evolution with recovery is preferred for the long-term operation of the bearings [9]. This necessitates that the grease should continuously bleed oil at a certain rate over the entire service life, compensating for the loss of oil in the bearings [61]. As analyzed in the preceding section, the oil bleeding rate is primarily determined by the permeability of the thickener-rich layer. Fig. 17 and Fig. 18 depict the permeability and bleeding behaviors of greases with different thickener fractions and thickener fiber radii. Greases with

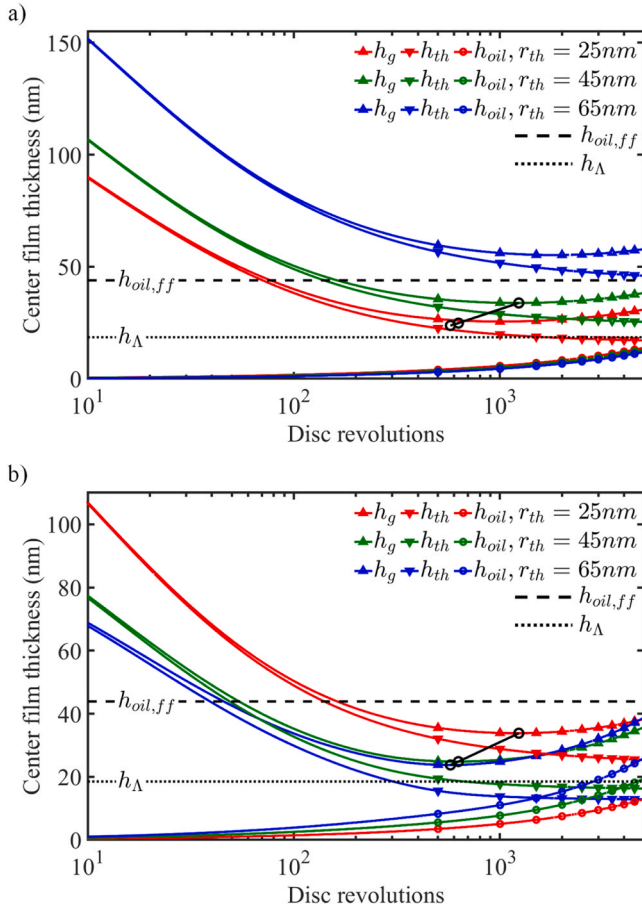


Fig. 16. Effects of the thickener on the film thickness evolution at contact center (a) Different thickener fractions when $r_{th} = 25\text{ nm}$, (b) Different thickener radii when $f_{th} = 10.5\%$ (for oil film thickness calculation, $A = 1.5$, $B = 1.67$).

lower thickener fractions or larger thickener radii exhibit higher permeability (see Fig. 17), and consequently, higher bleeding rates (see Fig. 18). This is the reason why PAO-Li-S exhibits a higher bleeding rate than PAO-Li-C, as shown in Fig. 15.

Furthermore, as revealed by Eqs.(16)–(18), the permeability is controlled by the thickener fraction and fiber radius. As shown in Fig. 17 (a), after approximately 2000 disc revolutions, the oil within the thickener-rich layer is completely displaced towards the side, resulting in nearly identical thickener fractions and permeabilities for different fresh thickener fractions. Thus, the variation of the thickener fraction only has a minimal effect on the bleeding rate, see Fig. 18(a). In contrast, a grease with coarser fibers, as shown in Fig. 18(b) still keep bleeding oil in a relatively high rate at the end of the simulation interval of 5000 overrollings. This bleeding behavior agrees with the experimental results by Lin and Meehan [62], indicating that the grease with a larger fiber size has a higher bleeding rate.

Additionally, it is essential to avoid excessively high bleeding rates, as they can reduce the service life of greases [18]. The critical oil volume required to achieve fully flooded condition can be determined using the critical meniscus equation by Hamrock and Dowson [31] and the meniscus distance equation by Fischer et al. [21]. In this study, the critical oil volume for achieving fully flooded lubrication condition is determined to be $2.68 \times 10^{-3} \mu\text{L}/\text{mm}$. In comparison, the bled oil volume is $2.06 \times 10^{-3} \mu\text{L}/\text{mm}$ when $r_{th} = 65\text{ nm}$ in Fig. 18(b), which is close to the fully flooded lubrication volume. Ideally, after recovering to a sufficient film thickness, the oil bleeding rate should equal to the rate of lubricant loss, such as due to evaporation [63]. Therefore, the

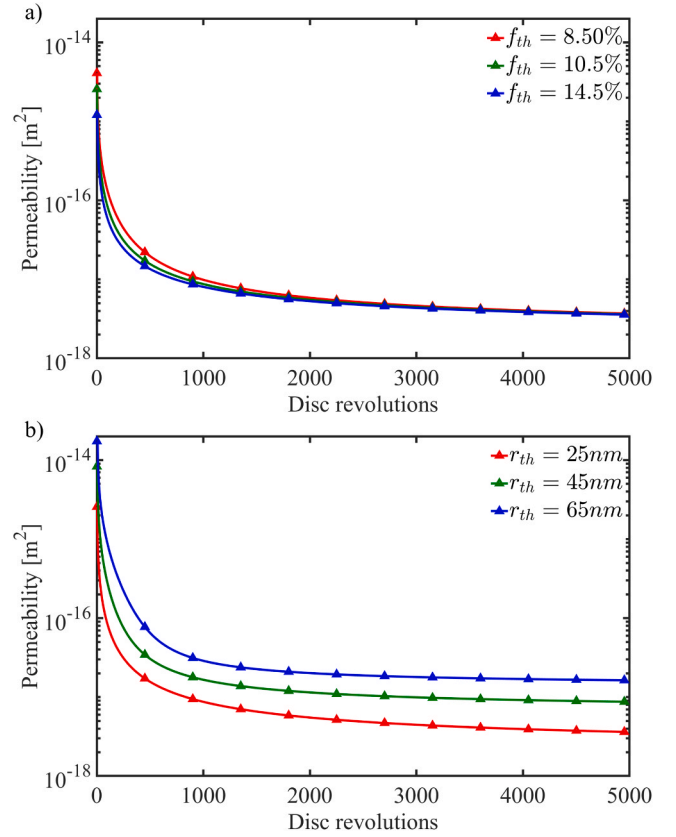


Fig. 17. Effects of the thickener on the permeability (a) Different thickener fractions when $r_{th} = 25\text{ nm}$, (b) Different thickener radii when $f_{th} = 10.5\%$.

numerical model presented in this study can offer guidance for selecting and designing of greases under various operating conditions.

5.3. Validation of film thickness evolution

5.3.1. Comparison with experiment

To validation the film thickness evolutions, the simulations employ the same lubrication and operational parameters as the experiments, as detailed in Table 1. The simulated and experimental central film thickness evolutions of two greases for approximately 5000 revolutions are shown in the Fig. 19. In this study, the first 10 measured film thickness data are neglected due to the high scattering in the churning phase. Because the PAO-Li-S has a lower thickener fraction and a higher thickener radius than that of PAO-Li-C, the thickener contribution to the film thickness becomes less [60]. As a result, the film thickness for the PAO-Li-S is lower than that of the PAO-Li-C at the beginning of bleeding phase.

As shown in Fig. 19, the film thickness evolutions of two greases, both numerical and experimental results, show a starved film thickness with recovery. The recovery points for both greases agree with each other. The bled oil volume from PAO-Li-S is more than that from the PAO-Li-C (see Fig. 15), however, the oil film thickness of PAO-Li-S is lower than that of PAO-Li-C. This is attributed to the significant difference in the contact angle, as depicted in Fig. 8. As concluded by Fischer et al., a smaller contact angle can reduce the meniscus in front of the contact [21,22]. Therefore, the pressure building-up is limited and a lower film thickness is generated [19,20]. Comparing the h_{th} by prediction and h_{grease} by experiment, most measured grease film thicknesses exceed the predicted thickness of thickener-rich layer h_{th} , especially for zone II. This indicates grease film at least has two components. At the end of the experiment, the h_{th} formed by PAO-Li-C is $47.7 \pm 6.0\text{ nm}$. At same disc revolutions, the h_{th} by prediction is 45.9 nm , which gives a

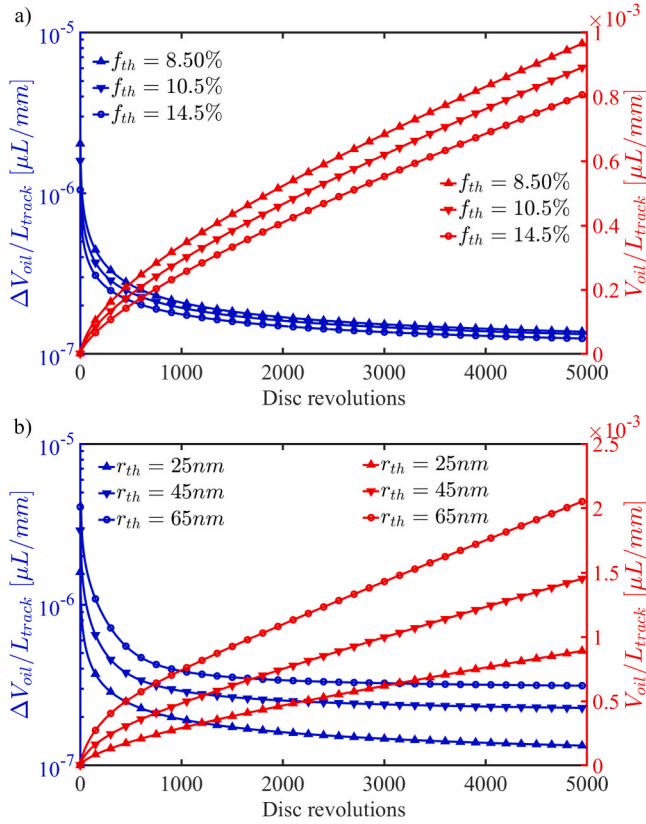


Fig. 18. Effects of the thickener on the oil bleeding behaviors (a) Different thickener fractions when $r_{th} = 25$ nm, (b) Different thickener radii when $f_{th} = 10.5\%$.

mean error about 1.8 nm. For PAO-Li-S, the predicted h_{th} is 12.1 nm. When compared to the experimental result, which is 13.2 ± 5.4 nm, the mean error is also less than 2 nm. Therefore, the predicted film evolutions have a reasonable agreement with the experimental results.

Moreover, the thickener-rich layer formed by PAO-Li-C gives an even higher thickness than the fully flooded film thickness $h_{oil,ff}$, which is indicated by the dashed line in Fig. 19(a). As a contrast, the h_{th} in the Fig. 19(b) using PAO-Li-S goes below the fully flooded thickness after about 40 disc revolutions and is below h_A (indicated by the dot line) after about 300 revolutions. In this case, the thickener-rich layer is not sufficient to separate the contacting surfaces and the formation of oil film becomes critical. Thus, this model provides deeper insights into grease lubrication by offering a more profound understanding of the evolution of a two-component film thickness.

5.3.2. Deviation analysis

Two deviations between the experimental and simulation results can be observed, especially for the PAO-Li-C. At the beginning of the zone I, the experiment results have a higher thickness than the predicted thickness. This derivation can be attributed to the particle effects, as shown in Fig. 20 and Fig. 21. At the beginning of testing, see Fig. 20(b) and Fig. 21(b), there are thickener particles passing the contact, which are also shown by [17,64,65]. Comparing with the PAO-Li-C, less particles are visible in the contact zone for PAO-Li-S. Therefore, a good agreement can be observed between predicted and experimental film thicknesses in Fig. 19(b) zone I. With the proceeding of the experiments (see $n \approx 1000$ and $n \approx 2000$ in Fig. 20 and Fig. 21), the particles are absent. Possible reasons are that the particles can facilitate the thickener-rich layer or disperse in the bled oil [66]. Therefore, the predicted film thickness can have a better agreement with the experiments at these stages.

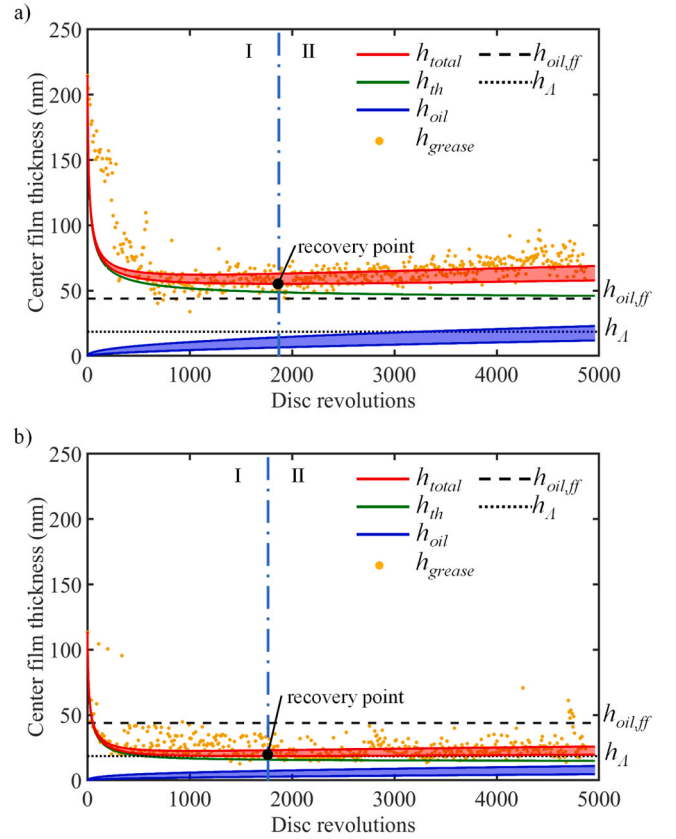


Fig. 19. Predicted and experimental film thickness evolutions for two different greases (a) PAO-Li-C (b) PAO-Li-S.

Another derivation comes at the end of zone II in Fig. 19, where the thicknesses of thickener-rich layer agree well with tested results. Therefore, it is believed that the predicted oil film thicknesses are lower than the oil film thicknesses observed in the experiments. This discrepancy can be attributed to other lubricant supply mechanisms, such as greases soften under the shear degradation [39], leading to a similar flow ability as the base oil [18]. Meanwhile, it is possible that other driving forces of bleeding exist, which are not included in this study, such as the centrifugal force [4] or other unidentified sources [15]. These effects can be the research directions for further work.

6. Conclusions

In grease-lubricated rolling contacts, maintaining a sufficient lubricating film thickness is crucial to reduce the risk of asperity contact. An accurate prediction of grease film formation requires models that can account for the transient formation of both the thickener-rich layer and the oil film. Therefore, this study introduces a porous thin layer model to simulate the transient formation of the thickener-rich layer and a multiphase bleeding model to simulate the oil bleeding. Using these models, film thickness evolutions are explored under various conditions, leading to the following key findings:

- 1) The thickener-rich layer primarily forms due to the lateral flow of the base oil within the thickener-rich layer under contact pressure. This lateral flow increases the thickener fraction, causing the thickener-rich layer to become progressively more viscous and behave like a solid layer over time.
- 2) The oil bleeding from the grease reservoirs is a result of the capillary pressure generated in the thickener-rich layer. The significant reduction in the permeability of the thickener-rich layer during

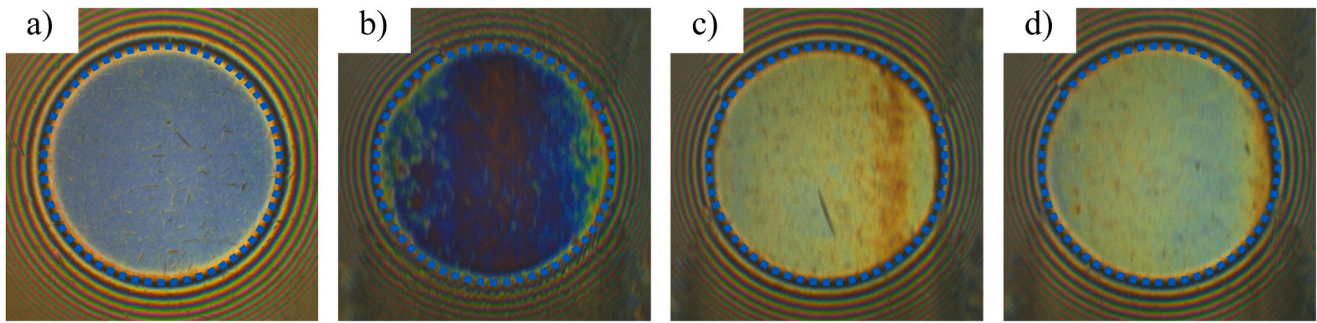


Fig. 20. Thickener particles in the contact zone for PAO-Li-C (a) clean contact zone (b) $n \approx 100$ (c) $n \approx 1000$ (d) $n \approx 2000$.

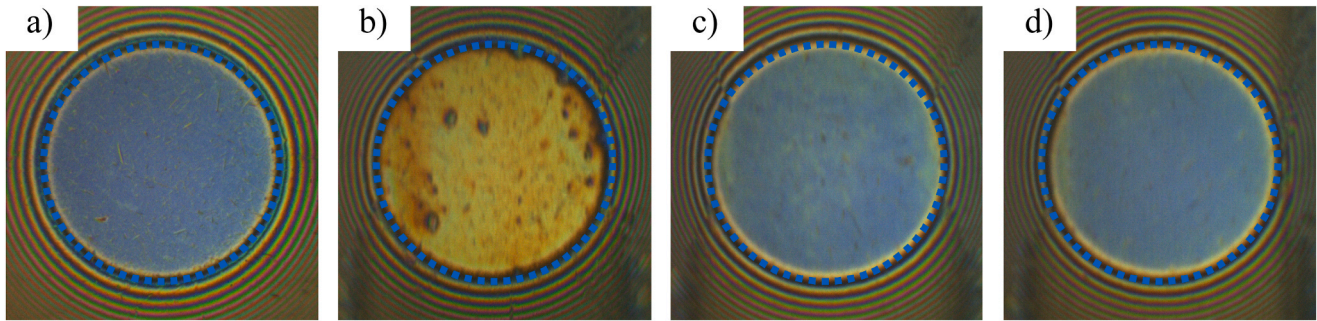


Fig. 21. Thickener particles in the contact zone for PAO-Li-S (a) clean contact zone (b) $n \approx 100$ (c) $n \approx 1000$ (d) $n \approx 2000$.

operation makes it a dominant parameter influencing the rate of oil bleeding.

- 3) Greases with a lower thickener fraction and coarser fiber structures are less effective at forming a thickener-rich layer. However, these greases exhibit a higher bleeding rate due to increased permeability, while variations in thickener fraction have only a minimal effect on the bleeding rate.
- 4) The proposed models successfully describe bleeding behavior and the transient film formation in grease-lubricated rolling contacts. The predicted grease film thickness evolutions, obtained by superposition of the oil film and the thickener-rich layer, demonstrate reasonable agreement with experimental results.

To the best of our knowledge, this introduced model represents the first comprehensive approach for predicting film thickness evolution in grease-lubricated rolling contacts, considering the transient formation of both oil film and thickener-rich layer. In future work, it is necessary to conduct further experimental validations to ensure the accuracy and applicability of this model for various operating conditions and greases. Also, the transferability to the entire bearing should be considered.

Novelty Statement

On behalf of my co-authors, I declare that the work described in the manuscript.

“Prediction of bleeding behavior and film thickness evolution in grease lubricated rolling contacts” has not been published previously in any other journal and neither is under consideration for publication in any other journal.

CRediT authorship contribution statement

Shuo Zhang: Writing – original draft, Writing – review & editing,

Visualization, Validation, Software, Methodology, Investigation. **Benjamin Klinghart:** Writing – review & editing, Validation, Methodology, Investigation. **Georg Jacobs:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Stephan von Goedel:** Writing – review & editing, Conceptualization. **Florian König:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization..

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Federal Ministry
for Economic Affairs
and Climate Action .

Appendix

Sensitivity check of porous thin layer model.

In this work, the thickness of the thickener-rich layer (h_{th}) is assumed to be twice that of the thickener-rich layer on the disc. The sensitivity check is carried out for various percentages of thickener-rich layer on disc and ball surface. The results are shown in the Fig. A1. The film thickness of the thickener-rich layer when using this 50%–50% assumption shows basically same thickness evolution with 75%–25% condition, and only a slight difference can be observed with the 90%–10% condition. A small difference between 50%–50% condition and 100%–0% condition can be observed. However, the film thicknesses become nearly identical after the first 1000 overrollings. Therefore, although current model is developed based on the 50%–50% assumption, it is still valid for a wide range of thickener-rich layer formation condition.

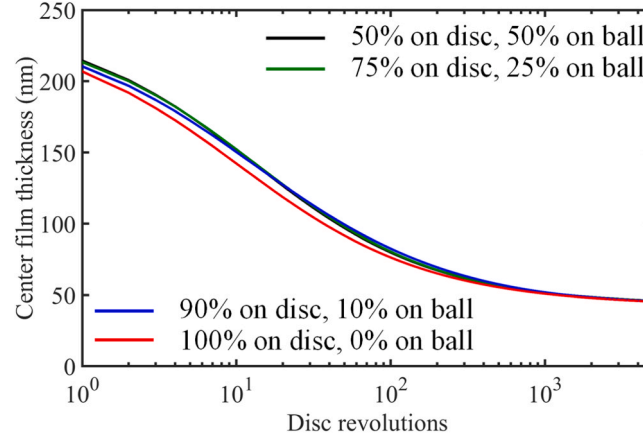


Fig. A1. Sensitivity analysis of the equal thickness share assumption.

Simulation domain and mesh independent study of PMB model.

The simulation domain of the PMB model can be determined by assuming the grease volume on the disc before churning and after churning is same. The grease track is assumed as a curved surface with same radius with ball, as shown in Fig. A2. The volume of the grease can be calculated as:

$$w_{model} \cdot h_{model} = A_1 + A_2 + A_3 + A_4 \quad (A1)$$

where A_1 , A_2 , A_3 and A_4 are the area of zones marked in Fig.A2 and can be calculated by Eqs.(A2)-(A8).

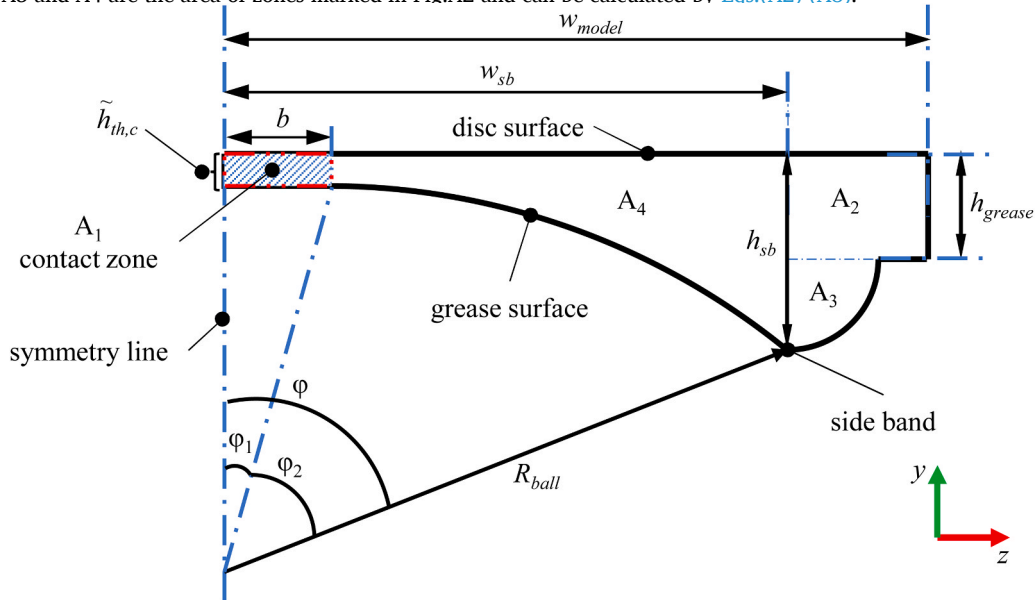


Fig. A2. Schematic representation of simulation domain.

$$A_1 = b \times \tilde{h}_{th,c} \quad (A2)$$

$$A_2 = (w_{model} - w_{sb}) \times h_{grease} \quad (A3)$$

$$A_3 = \frac{1}{4} \pi (h_{sb} - h_{grease})^2 \quad (A4)$$

$$A_4 = \frac{1}{2}w_{sb} \left[h_{sb} + \left(R_{ball} + \tilde{h}_{th,c} - \delta_{Hz} \right) \right] - \frac{\varphi_2}{2\pi} \pi R_{ball}^2 - \frac{1}{2}b(R_{ball} - \delta_{Hz}) - A_1 \quad (A5)$$

$$\varphi_2 = \arctan \varphi - \arctan \varphi_1 \quad (A6)$$

$$\tan \varphi = \frac{w_{sb}}{\left(R_{ball} + \tilde{h}_{th,c} - \delta_{Hz} \right) - h_{sb}} \quad (A7)$$

$$\tan \varphi_1 = \frac{b}{R_{ball} + \tilde{h}_{th,c} - \delta_{Hz}} \quad (A8)$$

The only unknown parameter in above equations is the position of the side band w_{sb} , which can be determined by substituting Eq. A2~A8 into Eq. A1.

To estimate the effects of the mesh number and the model size, the model-independent studies are conducted, as depicted in Fig. A3 and A4. After balancing the simulation time and the accuracy, the mesh model marked with red circles is selected. Fig. A4

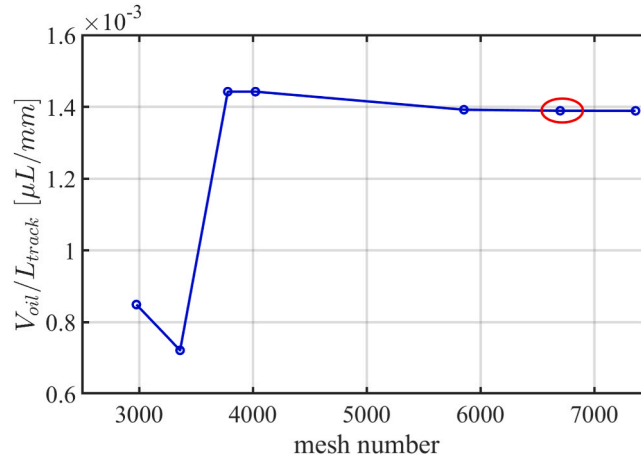


Fig. A3. Mesh number independent study ($u_{disc} = 0.05$ m/s, $w_{grease} = 3$ mm).

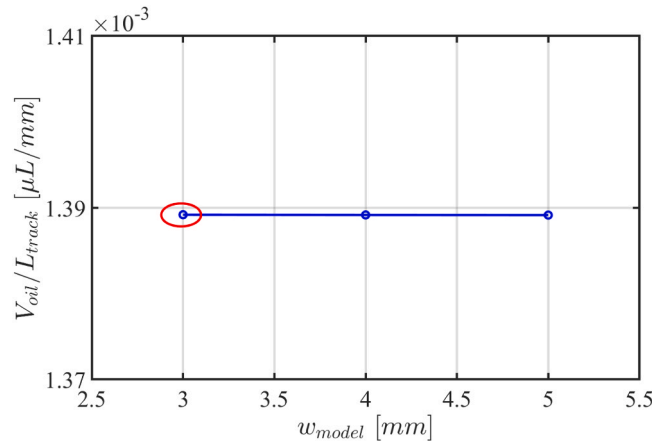


Fig. A4. Model size independent study ($u_{disc} = 0.05$ m/s, mesh cell = 6700).

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